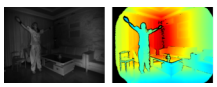
FIVE SENSORS.
ONE SMALL MODEL.

Can a machine recognise everyday activities without an ordinary camera?

one training clip (jumping jacks) as each sensor sees it

IR + DEPTH



sees the body and the scene

THERMAL



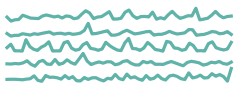
sees heat: bodies and objects

SKELETON



tracks 17 joints

WRIST IMU



feels the motion of the hands

RADAR



keeps the order of a recording

FIVE SPECIALISTS

one small model per sensor: three video, one skeleton + motion ensemble, one motion forest

body · hands · heat · posture · motion

FUSION + CONTEXT

combine the five votes, then use the natural order of a household routine

40 EVERYDAY ACTIVITIES

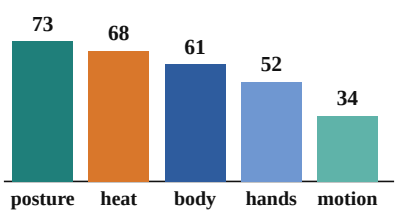
wash face, pour a drink, sweep the floor, jog, take medicine, ...

96 MB	0.877	UNSEEN	0 / 405
one file, every weight	private score, 14th of 307	people at test time	predictions changed on 5 GPUs

THREE THINGS WE MEASURED

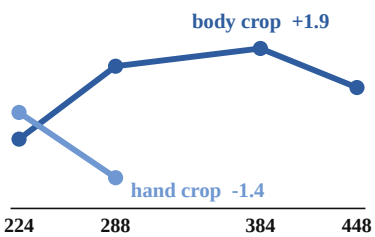
01 Different sensors rescue different mistakes

clips (of 2,700) that only ONE sensor gets right



some sensor is right on 2,399 clips;
remove any one and the system loses 11 to 48

02 More pixels are not always more information



the body fills its crop, so detail is real;
the hand is tiny, so extra pixels are interpolation

03 More model is not better generalisation

more capacity	✗
more training tricks	✗
more test-time tricks	✗
more input resolution	✓
a new sensor view	✓
better fusion	✓
recording structure	✓

every lever measured against the same statistical bar; the failures are the map

Team AGX · Atharv, BITS Pilani, K K Birla Goa Campus · CUHK-X Challenge 2026, Small Model Track

technical report follows